Detecting AI Model Substitution:

A Blockchain-Based Behavioral Verification System

I'm excited to share a project addressing a critical challenge in AI security: detecting when an AI model has been secretly replaced or modified.

The Problem:

As AI agents become more prevalent in critical applications, there's a growing risk of "model substitution attacks" where a legitimate AI model is swapped with a cheaper, less capable, or malicious alternative without users' knowledge.

The Solution:

I've built an AI Agent Fingerprinting System that combines blockchain immutability with behavioral trait verification to detect model drift and substitution attacks.

Key Features:

- 🔐 Cryptographic fingerprinting stored on Ethereum blockchain
- 🧠 Behavioral trait hashing using standardized test suites
- 🔍 Drift detection comparing current vs. registered behavioral patterns
- ⚡ Framework-agnostic architecture (React, Vue, Angular compatible)
- 🌐 Live on Sepolia testnet

Technical Approach:

The system uses a two-layer verification approach:

1. Static Fingerprinting: Cryptographic hashes of model identifiers stored on-chain with EIP-712 signatures
2. Behavioral Verification: Keccak256 hashes of response patterns across reasoning, ethics, code generation, and error handling tests

When behavioral traits drift from the registered baseline, it flags potential model substitution, fine-tuning, or configuration changes.

Why This Matters:

As enterprises deploy AI agents for customer service, financial analysis, and critical decision-making, they need assurance that the AI model they contracted for is actually the one serving their users. This system provides verifiable proof and continuous monitoring.

Tech Stack:
Solidity 0.8.20, Ethereum/Sepolia, ethers.js, React, TypeScript, Hardhat

The code is designed to be framework-agnostic at the core, with only the UI layer being framework-specific. This allows organizations using Vue, Angular, or other frameworks to easily adopt the verification system.

Interested in AI security, blockchain verification, or model integrity? I'd love to connect and discuss further applications of this approach.

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